

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester IV Examination, 2013-14

Botany
Paper: BOB-401 — Phanerogams

Time: Three hours

Maximum Marks: 70

Instructions: Answer five questions in all, selecting two each from Sections A and B and Question 9 of Section-C which is compulsory. The figures in the right-hand margin indicate marks.

Section-A

Question 1. Describe the principles of Bentham and Hooker's system of classification and discuss briefly its outline, merits and demerits. [15]

Question 2. Differentiate between any two of the following pairs of families: [15]

- (a) Rosaceae and Myrtaceae
- (b) Apocynaceae and Asclepiadaceae
- (c) Ranunculaceae and Papaveraceae

Question 3. Give distinguishing characters and economic importance of the family Solanaceae. [15]

Question 4. Write short notes on any two of the following with special reference to anatomical anomalies: [15]

- (a) Bignonia stem
- (b) Boerhaavia stem
- (c) Dracaena stem

Section-B

Question 5. Write short notes on any two of the following: [15]

- (a) Cycas rachis
- (b) Dwarf shoot of Pinus
- (c) Embryogeny in Pinus

Question 6. Describe the female gametophyte development in angiosperms. [15]

Question 7. Write short notes on any two of the following: [2 × 7.5 = 15]

- (a) Microsporogenesis in Angiosperms
- (b) Nuclear endosperm in Angiosperms
- (c) Double fertilization

Question 8. Describe briefly the structure and reproduction of *Ephedra*. [15]

Section-C

Question 9. Answer the following: [10]

- (a) Name a family in which parietal placentation is found.
- (b) Give the botanical name of an important medicinal plant of Lamiaceae.
- (c) Mention the botanical name and family of Kalmegh.
- (d) Give the botanical name and family of the plant which produces Quinine.
- (e) What is spikelet?
- (f) Write the name of a plant in which inverted cortical bundles are found.
- (g) Name the plant having amphivasal vascular bundles.
- (h) What is Chalazogamy?
- (i) Define polyspermy.
- (j) What are bordered pits?

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